Chase Golden Hockey Stick Graph

The Inconvenient Truth

In the age of climate awareness, data visualization is now a compelling storytelling tool to sway public opinion and stimulate action around climate change. The hockey stick graph exemplified in the 2006 documentary An Inconvenient Truth is one of the most recognizable examples. The hockey stick graph compactly illustrates the relationship among concentrations of CO_2 in the atmosphere, climate history, and global temperatures over a multi-millennial timeframe, culminating in an unmistakable spike for CO_2 levels and recent centuries of temperature anomalies, immediately making the argument for the alarm and urgency over the climate crisis accessible to massive audiences around the globe.

The goal of the work presented here was to validate and build upon the influential plot that Al Gore took the world through by cleaning data about CO_2 concentrations and temperature anomalies through Tableau Prep, and plotting visualizations around those datasets in Tableau Desktop. I aimed to replicate the visually arresting arc presented in An Inconvenient Truth and apply the modern conventions of dashboarding to allow greater clarity, interactivity, and analytical depth. The work culminated in an interactive, exploratory dashboard that allows users to interact with previously visible and obscured timeframes and trends spanning millennia, while highlighting the disturbing increase in greenhouse gas and temperature changes that have occurred in recent years.

Figure 1: Al Gore's Graph of the Inconvenient Truth

1 Recreating Al Gore's Graph

The original hockey stick graph made famous by Al Gore in An Inconvenient Truth has such a visual wow factor: for thousands of years, both levels of atmospheric CO_2 and global temperature oscillated between a statistically normal range, until the late 1800s when they suddenly began to jump upward. The dramatic picture of carbon emissions helped real audiences understand how profoundly different the climate was.

To create a similar effect, I started preparing the datasets in Tableau Prep. The datasets included ice core measurements of CO_2 concentrations and temperature anomalies going back 700,000 years at 50-year intervals. I adhered to traditional best practice by calculating new fields to transform the Years Before Present into a traditional AD/BC timeline, to allow the x-axis of the visualization the ability to show time from 700,000 BCE all the way to the present day.

With my data prepped in Tableau Desktop, I produced a dual-axis line graph with CO_2 and temperature overlaid over time. To evoke the same sharp rise as Gore's, I smoothed the lines, I used bold and high-contrast colors, and I used a black background with a dark gray grid to help emphasize the steep incline in our recent history and to discourage interest elsewhere in the visual.

Furthermore, I picked out the key inflection points, tracked the recent time, added reference bands, and colored gradients to accentuate areas of significance. I additionally stayed true to the original graph's essence of simplicity and emphasis yet added some additional contemporary dashboard elements to allow for exploration and context while analyzing the same data. The final visual maintains the emotional weight and visual presence of the original, but now offers much more ability for exploration and insight.

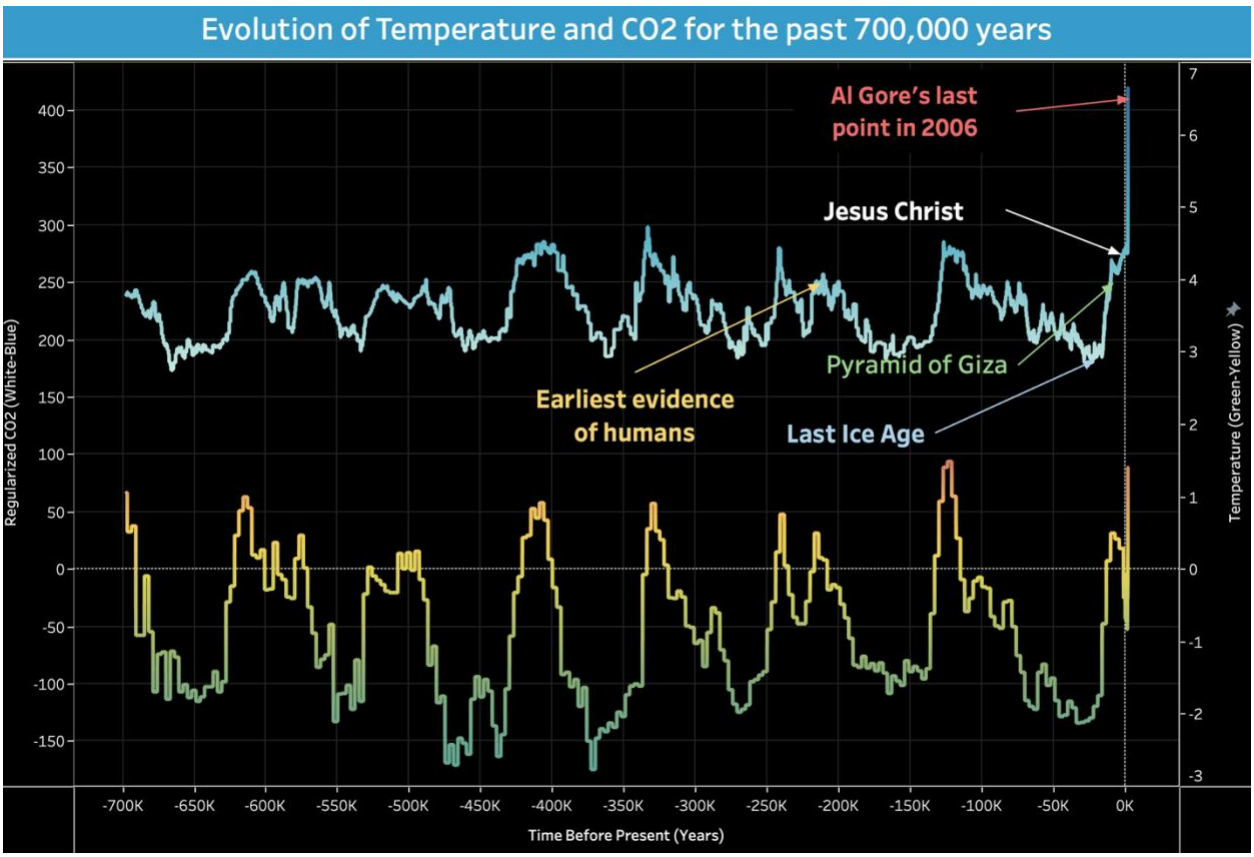


Figure 2: Final Dashboard Inspired by Al Gore's Hockey Stick Graph

2 Challenges Surpassed

Re-creating the hockey stick graphic was not a simple task. The first challenge presented itself in Tableau Prep. The dataset was clean, although I struggled trying to figure out how to correctly join the CO_2 and temperature files. One error I made was that I did not realize I had to set the join key field, which led to some confusion and delay on my end. After I fixed that issue, I was able to join the datasets by Year and, in the end, combine both metrics into one flow. I also had some trouble figuring out how to set up calculated fields and how to then take those definition fields and label each step as I would in Prep, but with some help, I was able to get through the exercises.

Once I got to Tableau Desktop, my challenges became more visual, and the struggles were with the design elements. My first version of the graphic utilized colorful grid lines, and overall did not jump out at the audience. After I submitted a rough version to my professor, he commented that the grid was too much, and one of the first comments I got was that they saw the graph cut off the temperature line. My professor wanted me to make darker grid lines, change the range of the axes, temporal filter, and use color as a function of the level of data to direct the audience to the point of the visual experience.

After that, I worked on improving the visual presentation. I changed the grid lines' color to a darker shade, adjusted the dual axis so the temperature was not clipped at the bottom, and recolored the CO_2 and temperature lines to a more common convention. One of the more challenging items was getting the annotations and highlighted temperature bands to display properly. I had a hard time finding the "Add Reference Band" button, and when I did find the option and used it, the reference band did not behave as I had expected. After some work, I was able to label specific references to, and to also add the temperature anomaly color bands.

Finally, my professor suggested transforming the sheet into a dashboard for improved usability and capabilities. Having a singular dashboard layout required some adjustments to the size of the chart, as well as removing blank margins, and fitting everything snugly to maintain interactivity. This stage also involved adding title banners, image elements, and refining legends and tooltips for clarity.

All in all, the main takeaway has been how exploratory and granular visual storytelling is when attempting to match the impact that an iconic visualization has as a persuasive landscape. I spent a lot of time troubleshooting uncomfortable formats, layouts, and interactions with Tableau that were not obvious, but ultimately produced a dashboard that felt authentic to the original visualization of the Inconvenient Truth, while still being able to add nuance and academic platform.

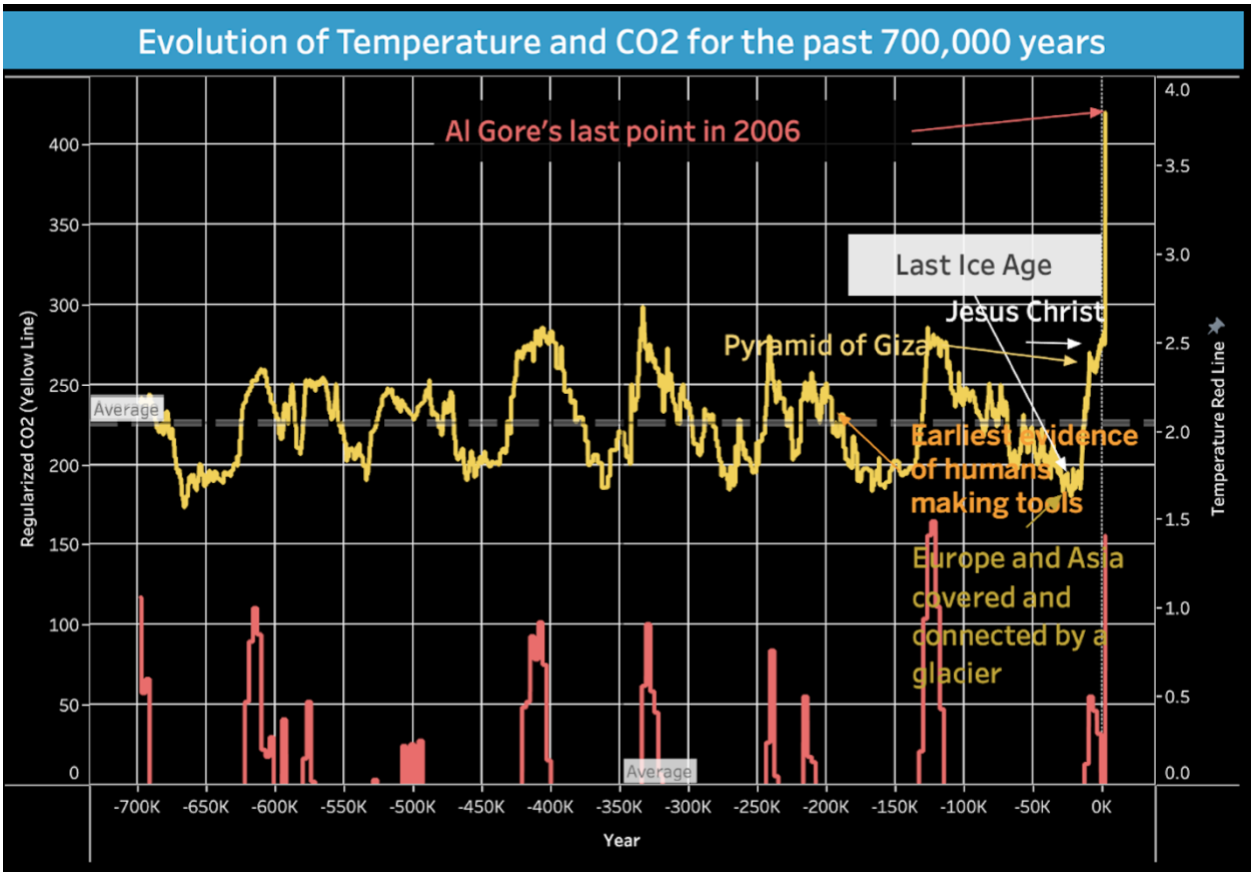


Figure 3: Rough Draft of the Dashboard with Heavy Gridlines and Cut-Off Axes

3 Unmet Goals and What I'd Do with More Time

Although I created a reasonable approximation of the hockey stick graph and replicated the key components of Al Gore's presentation, I did miss some aspirational features, primarily because of my limited skill in Tableau and the complexity of the assignment.

One of the goals I initially set was to create interactive, colored temperature bands with a gradient-like CO_2 background, like my professor's introduction. I experimented with reference bands and attempted different layering strategies, including changing the order of visual layering on the marks card, but I could not achieve the smooth, red/blue fade for CO_2 or the bold green bands for the highlighted temperature bands. At one point, I attempted to use references and annotations that approximated the feel of what I wanted, but it ultimately was not visually as appealing as the sample. Given more time or, better yet, more experience in dual-axis chart formatting or background images, I think I would have been able to create a more precise evening format.

Adding a filter that lets the user isolate eras in time was another suggestion made by my professor. Ultimately, I added the dimension Year, but I was not as confident adding a filter of the period without knowing how it would affect the layout, confuse the dual-axis scale, or perhaps break it. If I had more time, I would want to explore the ability to use a range slider or drop-down filter to refine the period and explore the data interactively without modifying the graph build.

Finally, while the dashboard now sits inside of a well-organized dashboard container with annotations and a title, I did not incorporate image icons, storytelling banners, or links to outside materials, like my professor did in their version. These enhancements could also add value to the overall polish of the design and storytelling value of the design and would most definitely be on my list of improvements if I were submitting it to a public portfolio or presenting it in a job interview.

4 What I Recreated from the Original Graph

The main goal of this exercise was to recreate the hockey stick graph from An Inconvenient Truth, which shows a correlation over time between atmospheric CO_2 levels and global temperature anomalies. In the end, I was able to capture the essence and the visual story from the original graph. This included:

- A dual-axis line graph with Regularized_CO2 and Regularized_Temperature anomalies over a 25,000-period
- The x-axis was constructed with logical, forward-looking year labels created from the original years before present in Tableau Prep.
- A clear visual spike in CO_2 and temperature beginning around the advent of the industrial era, which emulated the very dramatic spike on the upward blade of the hockey stick graph.
- Color-coded lines for the two measures with gradients and emphasis for the dramatic recent rise for each variable.
- Interactive tooltips that allow the user to check year-specific data for CO_2 and temperature.

I also added several modern visual enhancement features. I created a dark background with some dramatic contrast, created temperature reference bands to give it some context, and added annotations at critical years to enhance the visuals. These stylistic choices are like my professor's original dashboard design and emphasize the intended story of the original graph: the acceleration of climate change on planet Earth at an advanced rate in modern times.

5 Creative and Unique Additions

While I intended to faithfully reproduce An Inconvenient Truth's hockey stick graph, I wanted to take the opportunity to add design elements as well as creative decisions that were consistent with my sense of analysis but also made it clearer to an academic audience.

One of the most important design elements I made was to choose a dark gray grid as well as a dark gray background rather than a standard white Tableau canvas grid. Based on my professor’s feedback, he determined the original white background grid was too visually harsh, as the common premise of any form of a graph is to display data clearly and accurately. Switching the background to dark gray softened my graph, which allowed the CO_2 and temperature lines to pop better off the graph grid. I also used the intensity of the colored line. CO_2 was set to a bright red line, while the temperature line was a solid, glowing yellow line. Both colors intuitively presented the subject's heat and urgency while still providing the necessary high contrast to sustain the ability to read the graph quickly.

I decided to use a dual-axis line chart so that both CO_2 and temperature could be plotted at the same time on the same time axis while maintaining a distinct vertical scale. This design allows the viewer to see the dynamic correlation between the two variables through centuries

without one overpowering the other. To make the view understandable in the dual axis, I modified the scales and line thickness and made sure the axes were labelled clearly.

I made it better in the areas of annotation and formatting. I added callouts on significant years to indicate when industrialization began to ramp up CO_2 and when the most recent temperature increase occurred. The callouts show more than the data; they mark historical turning points in the evolution of climate change awareness. I also embedded the visual into a suitable dashboard container rather than a single sheet. The dashboard embedded the visual and provided a lot of flexibility in the layout, before it was presentation-ready for design.

Although some of these creative elements stemmed from my professor's example, I was intentional and purposeful with each design decision. They reflect my evolving eye for good visual storytelling, as well as my improving level of proficiency in Tableau.

6 Storytelling and Insight

The dashboard is intended to tell a visually compelling story of the historical relationship of atmospheric CO_2 levels with global temperature anomalies, one that approximates the effects of the original hockey stick graph while displaying recent data and incorporating interactivity. My visualization provides a clear, direct, and urgent message: the relationship between rising CO_2 levels and rising global temperatures is not only undeniable but accelerating.

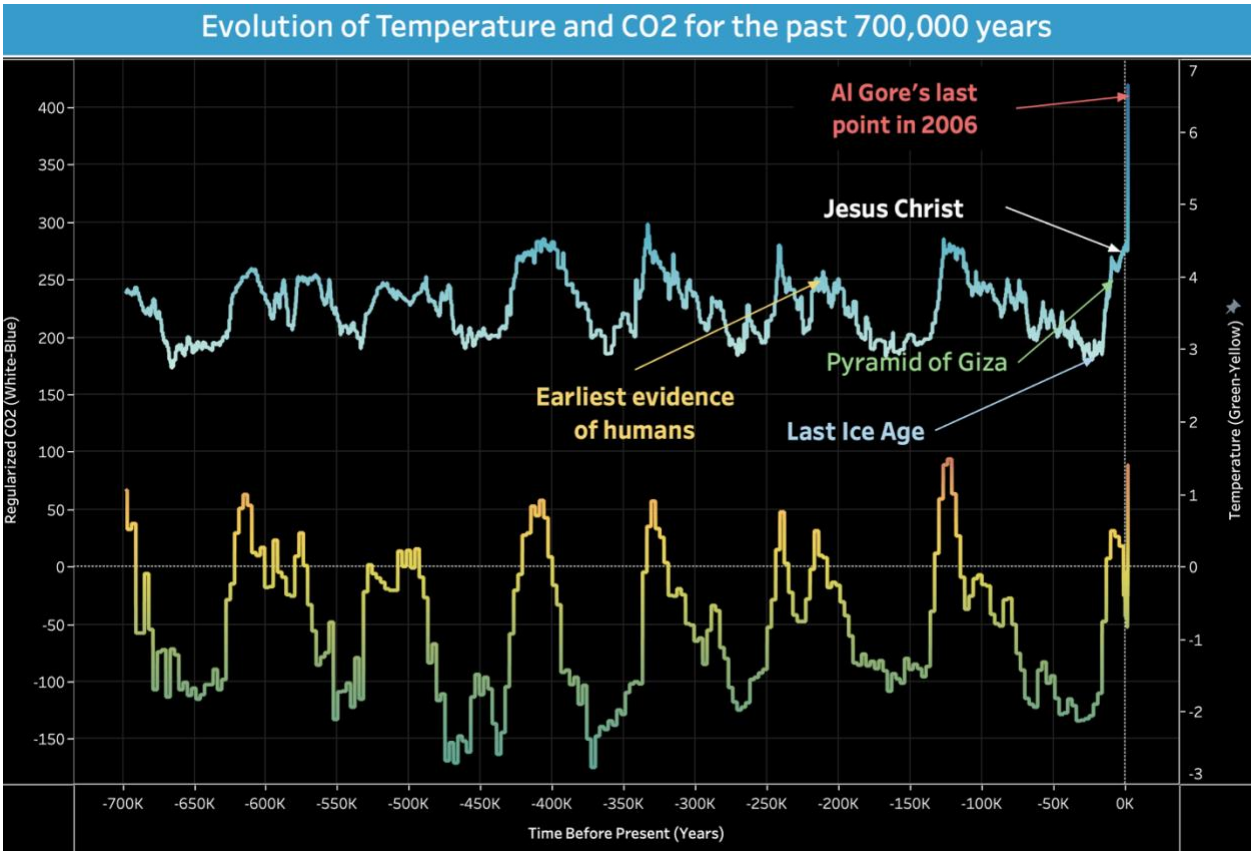
The dashboard combines CO_2 and Temperature trends on a dual-axis time series that effectively shows a basic trend over centuries, followed by a sharp and sudden increase beginning around the year -1850. This essentially highlights the inflection point of the human impact on climate and continues to depict the idea of anthropogenic climate change. The ending point is a dramatic apex that emphasizes the urgency of the climate crisis.

The format assists in this comprehension for participants in survey research and for researchers. The graphic angle of both lines produces the blade of the hockey stick, which is a metaphor that is relevant and easy to understand. Color can tell part of a story too here: the red color of CO_2 suggests heat, danger, and act quickly, and the bright yellow temp suggests a level of intensity. Altogether, using a dark background, these colors tell a warning story without being overwhelming.

Labels are situated on the timeline to clarify the user’s understanding and serve as historical reference points for the data. The ability to hover over data, enabling users to see how temperature and CO_2 changed in the different periods, also enhances the interactivity of the dashboard. In upcoming iterations, I can include a temporal filter so users can once again visually zoom into their selected time frames, while still allowing for the narrative flexibility.

This dashboard visually leads users through both a chronological journey as well as a temporal progression. It is refreshing modeling and language used are not overly complicated with complex reasoning or terms, but rather, the clear and abiding observational trend of the climate data is presented simply, authentically, and in a data-driven manner, which is intended to evoke emotion. This is a great indicator that the resulting visualization is both academically credible and targets the intended emotional impact on users in a quite visceral way. At the same time, an acceptable portrayal of the original "An Inconvenient Truth," the dashboard stands fundamentally on its own as a compelling data story, but roughly as impactful as some degree of constructive criticism to truly represent the whole story of climate change.

Appendix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Jesus Christ
Pyramid of Giza
Earliest evidence of humans
Last Ice Age
Regularized CO2 (White-Blue)
Temperature (Green-Yellow)
Time Before Present (Years)
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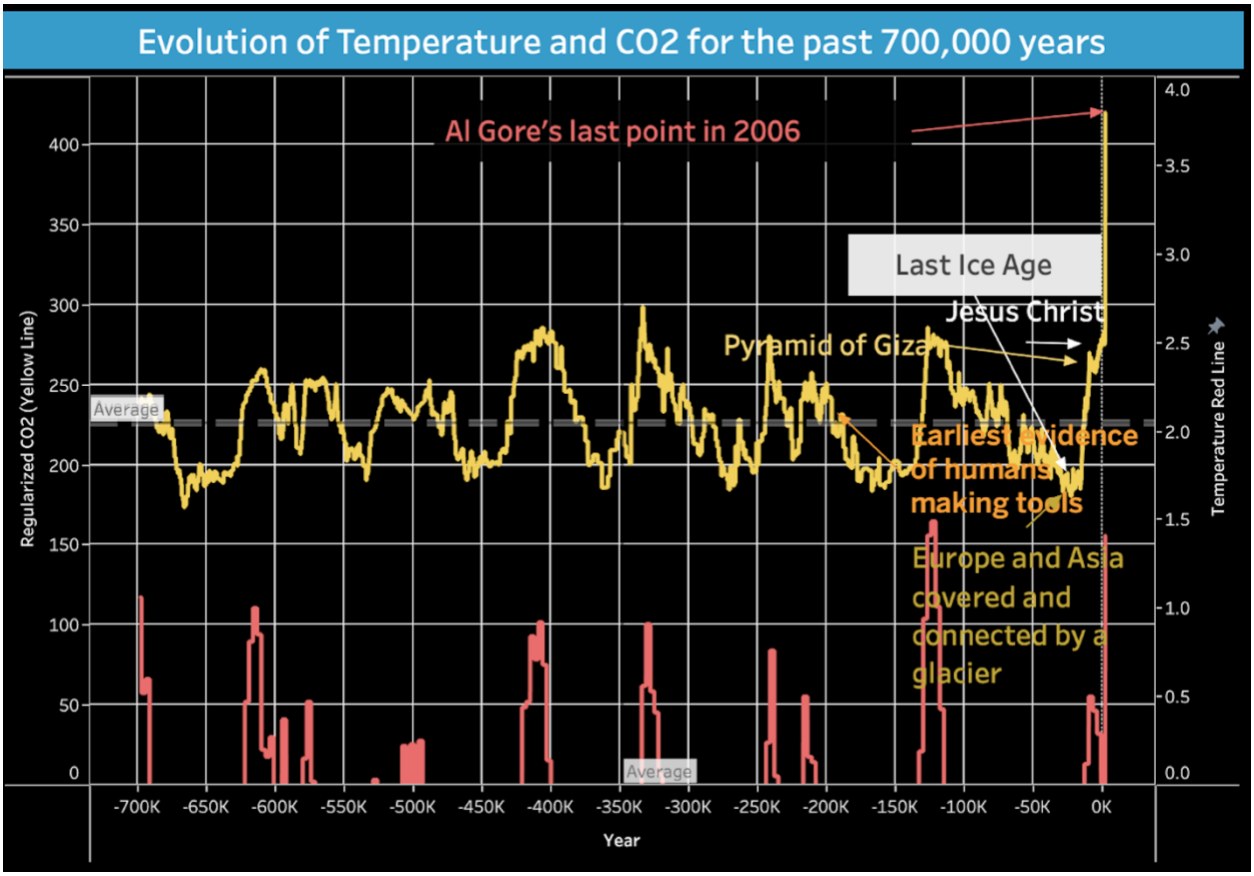
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Average
Average
Regularized CO2 (Yellow Line)
Temperature Red Line
Year
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